DEBONEEL KUNDU PARTHO

Aspiring Research Assistant - Climate-Resilient Irrigation, GIS & Water Resources Modeling

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LinkedIn | Google Scholar | ResearchGate | Portfolio

PROFILE

Agricultural engineering graduate (Irrigation & Water Management) with four Q1 journal publications in climate resilient irrigation, hydrology, and water quality modeling. Experienced in integrating CMIP6 climate ensembles, **GIS**-based spatial interpolation (kriging), crop-water simulation tools (**CropWat, AquaCrop, DSSAT**), and hydrological modeling systems (**SWAT+, HEC-HMS**) to evaluate climate change impacts on irrigation demand and watershed dynamics. Seeking a Research Assistant position to contribute technical and analytical expertise to a funded research program in water resources management or agricultural climate adaptation.

EDUCATION

B.Sc. in Agricultural Engineering - Irrigation & Water Management September 2026

Bangladesh Agricultural University, Mymensingh, Bangladesh

- CGPA 3.488 / 4.00 (8 semesters) - Ranked 2nd in Department of Irrigation and Water Management
- Undergraduate Thesis: *"Spatially Explicit Mapping of Future Irrigation Demand for Boro Rice in the Drought-Prone Rajshahi Division: A GIS Kriging Approach Using CMIP6 Ensembles, Bangladesh"* (under review)
- Supervisors: **Dr. Md. Touhidul Islam** (Associate Professor) and **Dr. A.K.M. Adham** (Professor), Dept. of IWM, BAU

RESEARCH EXPERIENCE

Research Assistant | *AgroSense and Water Innovation Lab, BAU* Jan 2024 – Present

- **Executed hydro-climatic modeling and spatial analysis** across Mymensingh district, leading CropWAT irrigation projections for primary crops (maize and potato, with supporting contributions to wheat) and mapping Brahmaputra River water quality from direct field sampling.
- **Guided and mentored 15+ research teams (50+ members)** across Rajshahi, Bogura, Rangpur, and Dinajpur districts, standardizing CropWAT, CMIP6 GCM ensemble downscaling, and GIS spatial interpolation workflows for sub-district irrigation planning.
- **Built a strong research foundation** through 3 seasons of hands-on AWD rice field trials - mastering agronomic data collection (LAI, grain yield, 1000-grain weight, panicle height, effective vs. non-effective panicle counts) and statistical data analysis - while mastering CMIP6 climate bias correction, Leave-One-Out Cross-Validation (LOOCV), statistical kriging, and spatial interpolation methods including IDW (power parameters p = 1, 2, 3).
- **WebAIS Project Member (2025–Present):** Collaborate with **BADC**, **BUET**, **BRRI**, and **TH Köln** to optimize stage-specific rice irrigation using **DSSAT** crop modeling across pilot sites in **Natore**, **Manikganj**, and **Satkhira.**

Field Research Associate | *Bangladesh Agricultural University* Nov 2024 – Present

- **Potato ET_0 Irrigation Modeling (2 Seasons):** Developed and validated an ET_0-based precision irrigation model for potato, reducing field water inputs while maintaining target yields under multiple simulated climate scenarios.
- **Potato Phytoremediation & Wastewater Reuse (1 Season):** Designed field trials evaluating potato (*Solanum tuberosum*) for phytoremediation, linking crop physiological stress responses to contaminant load reduction for safe wastewater reuse.

- **BRRI dhan88 Rice AWD Management (3 Seasons):** Implemented Alternate Wetting and Drying (AWD) field protocols across three seasons, guiding team members on soil water-table monitoring to minimize water use without compromising yield.

Research Intern - Agricultural and Fisheries Division | Center for Environmental and Geographic Information Services (CEGIS), Ministry of Water Resources Jun – Jul 2026

Selected for a 7-week technical internship at CEGIS (Ministry of Water Resources, Bangladesh) in the Agricultural and Fisheries Division, directly contributing to the North Rajshahi Irrigation Project by evaluating drought severity and water availability via the Agroecological Zones (AEZ) Database, processing hydro-meteorological datasets, producing soil and agricultural thematic maps in **ArcGIS Pro**, executing crop yield and water-requirement simulations using **FAO AquaCrop**, completing technical training in **SWAT (Soil & Water Assessment Tool)** and GIS/Remote Sensing for climate-smart irrigation, and co-authoring technical reports - earning formal commendation from the Executive Director (Acting) for research aptitude and multidisciplinary teamwork.

PEER-REVIEWED PUBLICATIONS (4 Q1 JOURNAL ARTICLES)

[1] Spatiotemporal Dynamics of Climate-Driven Irrigation Requirements for Maize in Mymensingh, Bangladesh: A GIS-Based Multi-Model GCM Ensemble Projection Approach

Results in Engineering (Elsevier) · **Q1 · IF 7.9** · DOI: 10.1016/j.rineng.2026.110970

[2] Future Water Management Strategies for Potato Irrigation Under Multiple Climate Change Scenarios Using Advanced CMIP6 Modeling in Subtropical Bangladesh

Journal of Agriculture and Food Research (Elsevier) · **Q1 · IF 6.2** · DOI: 10.1016/j.jafr.2025.102571

[3] Climate-Smart Irrigation Planning for Rabi Maize (Zea mays L.): CMIP6 Multi-Model Projections in North-Central Bangladesh

Journal of Agriculture and Food Research (Elsevier) · **Q1 · IF 6.2** · DOI: 10.1016/j.jafr.2025.102225

[4] Regional Irrigation Water Quality Index for the Old Brahmaputra River, Bangladesh: A Multivariate and GIS-Based Spatiotemporal Assessment

Results in Engineering (Elsevier) · **Q1 · IF 7.9** · DOI: 10.1016/j.rineng.2024.103667

Manuscript under review:

[1] Spatially Explicit Mapping of Future Irrigation Demand for Boro Rice in the Drought-Prone Rajshahi Division: A GIS Kriging Approach Using CMIP6 Ensembles, Bangladesh (undergraduate thesis manuscript)

[2] District-Scale Projections of Climate-Driven Divergence in Wheat Irrigation Requirements Under Shared Socioeconomic Pathways for Rangpur and Dinajpur, Bangladesh

TECHNICAL SKILLS

- **GIS & Spatial Analysis:** ArcGIS Pro, Arcmap, Spatial Interpolation & Kriging, Remote Sensing
- **Climate & Crop-Water Modeling:** CMIP6 multi-model GCM ensembles, CropWat, AquaCrop, DSSAT
- **Hydrosystems Modelling:** SWAT+, HEC-HMS
- **Programming & Data Analysis:** Python (NumPy, Pandas, Matplotlib, Seaborn), statistical and machine-learning modeling
- **Software:** AutoCAD, Microsoft Excel, Word, PowerPoint, HTML/CSS, Adobe Illustrator, Canva
- **Languages:** Bangla (native), English (professional working proficiency)

REFERENCES

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